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Citation: *Journal of Applied Physics* **104**, 094104 (2008); doi: 10.1063/1.3008006

View online: <http://dx.doi.org/10.1063/1.3008006>

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Charge injection and charge separation as revealed by dynamic space charge measurement in poly(propylene-ethylene) copolymer films

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(Received 25 May 2008; accepted 14 September 2008; published online 6 November 2008)

The time resolved fast pulsed electroacoustic method is used to investigate the space charge characteristics in an additive-free poly(propylene-ethylene) copolymer in film form. The 190 μm thick films, sandwiched by aluminum (at the lower electrode) and semiconducting material (at the upper electrode), were submitted to dc and low frequency ac stress for various durations at various temperatures. The results show that under dc stress, homocharges accumulate near the lower electrodes and heterocharges near the upper electrode at 25 °C whereas at 50 °C, besides the similar space charge behavior as at 25 °C in the vicinity of the electrodes, a region of symmetric separation of charges of opposite polarity is detected in the bulk. In addition, the time for the symmetric separation emergence under high voltage application depends on the field intensity, being shorter for higher field. Similar charging characteristics were also found at 50 °C under low frequency (5 mHz) square 50 kV/mm rms stress, with surprisingly periodical feature, while only injected charge was observed at 0.1 Hz square rms 50 kV/mm stress. Combining dc data with ac data, it is shown that the space charge build up results from two contributing processes: the space charge formation adjacent to the electrode is due to electronic carriers injection and drift, whereas the symmetric space charge in the bulk shows ionic transport characteristics, which can be well fitted with an ionic transport model. © 2008 American Institute of Physics. [DOI: 10.1063/1.3008006]

I. INTRODUCTION

The space charge behavior in polymeric insulating materials has been extensively investigated over the last decades.^{1–8} Under certain operating conditions, charge accumulation and migration in the bulk may occur during the whole insulation life. For a given material, the space charge behavior can be influenced by many factors, such as the applied field intensity, frequency, temperature, electrode material, and additives. In general, one can find more charge accumulating under dc stress than under ac stress. Due to the significant charge accumulation under dc stress, the local field can be greatly enhanced, which may lead to an accelerated material degradation or even to unexpected breakdown of the insulation.⁹ The amount of accumulating space charge under ac stress is much less than that under dc, but the periodical charge injection and extraction are believed to induce fatigue of the material by the charge frequently impacting on the molecular chains.¹⁰ It is therefore important, in both cases, to have a good description of charge generation and transport processes.

Several methods, such as the pulsed electroacoustic (PEA),¹¹ the pressure wave propagation,¹² and the thermal step¹³ techniques have been developed to measure space

charge density distribution profiles in solid dielectrics. This has been realized essentially under dc stress. The on-line charge measurement under ac stress needs more sophisticated devices, which may be the reason for finding much less reports under ac condition^{1,5,11,14} than under dc condition. In addition, there is no available information on the relationship of charge dynamics between ac and dc conditions in a given material. This paper investigates the space charge behavior in poly(propylene-ethylene) (PPE)-copolymers subjected to dc stress combined with charge evolution under low frequency square ac stress by using a fast acquisition PEA cell.

Due to its excellent physical and chemical properties, polypropylene (PP) is widely used as a dielectric material in power capacitors and cable wraps, as well as in layers and phase separators for rotating electrical equipments and transformers.^{15,16} As a recyclable material, some researches have been done on PP for power cable applications. As an example, the copolymerization of propylene with a small amount of ethylene could be a way to improve the mechanical properties of the PP (decrease in stiffness, in the same way as crosslinking does for polyethylene). However, only limited attention has been paid on the space charge characteristics on this kind of copolymer during ac stressing.

This paper reports on the space charge behavior in PPE subjected to dc fields in the range from 15 to 100 kV_{dc}/mm and to square ac stress of 50 kV/mm (rms) at frequencies of 5 mHz and 0.1 Hz. The experiments have been carried out at 25 and 50 °C for various stressing duration. Space charge density profile measurements were carried out by using the

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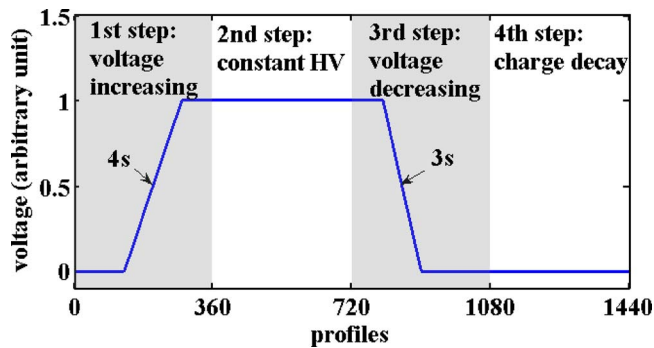


FIG. 1. (Color online) Typical space charge measurement protocol under dc stress. The frequency of acquisition is 1 profile/0.3 s in steps 1 and 3 and 1 profile/10 s in steps 2 and 4. Note that the X axis represents the profile number and not the time.

PEA method. By combining with the charge dynamics under ac stress, the origin of charge build up has been identified. In Sec. III of the paper, the effects of stress frequency and temperature on space charge behavior are discussed in details.

II. EXPERIMENTAL

The copolymer material used in this paper has been obtained from Goodfellow® in the form of 190 μm thick rolls. As scarce data were available on material properties, detailed physicochemical characterization has been carried out. The crystallinity of the material has been estimated as 33% by differential scanning calorimetry, considering for calculation the standard heat of melting for a 100% crystalline PP as 209 J/g.¹⁷ The isotacticity of the copolymer calculated by Infra-Red analysis¹⁸ is about 85%. The ethylene content as well as the isotacticity was calculated by solid state ¹³C-NMR.¹⁹ A value of 86.7% isotacticity was calculated from the methyl signals, consistently with IR data. The ethylene content was obtained from the methylene signals according to the method proposed by Cheng.²⁰ The obtained fraction of ethylene is 11.8 mol %.

All the experiments have been carried out in a temperature-controlled oven. Before testing, the temperature (25 or 50 °C) was equilibrated for 20 min. The high voltage (HV) was applied to the upper electrode (UE), which is made of a semiconducting material (polymer+carbon black). The lower electrode (LE), made of aluminum, was grounded. The voltage source is a 10 kV Trek® amplifier.

dc stressing was carried out under fields of 15, 30, 50, and 100 kV_{dc}/mm at 25 and 50 °C. The voltage was either positive or negative in order to investigate the effect of the polarity on the space charge behavior. The space charge profile measurement is divided into four steps as shown in Fig. 1. Ramped voltages were used in order to avoid sudden shocks to the film. The typical voltage increasing and decreasing rates are about 2.5 and 3.2 kV/s, respectively, for a field of 50 kV/mm. In order to detect possible high mobility charges, the profiles were acquired every 0.3 s during the first and third steps. In the second and fourth steps, profiles were acquired every 10 s. However, only representative PEA charge profiles are shown in this paper. In order to improve the signal-to-noise ratio of the PEA signal, all of the available signals in this paper have been averaged for 60 times by

oscilloscope (digital Lecroy LT374). The space charge behavior in two superposed PP films is also investigated.

The ac behavior has been tested under rms 50 kV_{ac}/mm square stress at 0.1 Hz and 5 mHz. During PEA measurement, at both stress frequency, every period is divided into about 60 profiles through controlling the PEA electric pulse frequency. More details about the setup can be found in Ref. 5.

III. RESULTS AND DISCUSSION

A. Space charge formation under dc stress

Figure 2 shows space charge profiles in PPE under dc stress at 25 and 50 °C, for positive and negative polarity on the UE. At 25 °C, a small amount of negative charge can be found in the insulation bulk almost immediately after a positive voltage is applied to the UE. In the course of time, negative charges accumulate near the LE with the negative charge density increasing in the bulk during 1 h stressing. No positive charges appear in this test [Fig. 2(a)]. Similarly, a small amount of positive charges arise in the sample bulk after a negative voltage is applied to the UE for about 1 min. After this, the positive charge density increases with the stressing time. No negative space charges arise in 1 h stressing [Fig. 2(b)].

In both cases, the dominant space charges after 1 h of stressing are homocharges adjacent to the LE. With the several hundreds of PEA profiles recorded, it is clear that the final dominant charge is injected from the LE. The charge detected adjacent to the UE most probably results from the drift of carriers injected at the LE and a limitation in the extraction flux at the UE. The fact that more charges are apparently injected from the LE than from the UE could be put in relation to the fact that the spatial resolution of the PEA method is lower when going from the LE to the UE. However, the dominant space charges on the long term appear to be those injected from the LE. Hence the difference in behavior has rather to be put in relation with the difference in the nature of electrodes (Al versus semiconducting material) and hence to different work functions. Focusing on the LE, one observes a larger charge quantity of injected negative charges than of positive ones. This is not really surprising as it has been reported²¹ that electrons and holes exhibit different injection rates for the same interface. Compared with the injected positive charges, the negative charges show higher injection rate for the same electrode and penetrate further into the bulk during the same period of measurement. This indicates that the negative charges have higher apparent mobility than positive charges at this temperature.

When the temperature is increased to 50 °C, a region with a symmetric distribution of charges appears in the bulk of the insulation. This region is materialized by a rectangle in Figs. 2(c) and 2(d). In the meantime, charges of opposite polarity to the UE are collecting near this electrode and near the LE. For example, in Fig. 2(c), a negative space charge grows up adjacent to the LE as well as with slower kinetics, adjacent to the UE. For a field of 50 kV/mm, the formation of the symmetric distribution of charges can be appreciated after about 30 s, the density of the two kinds of opposite

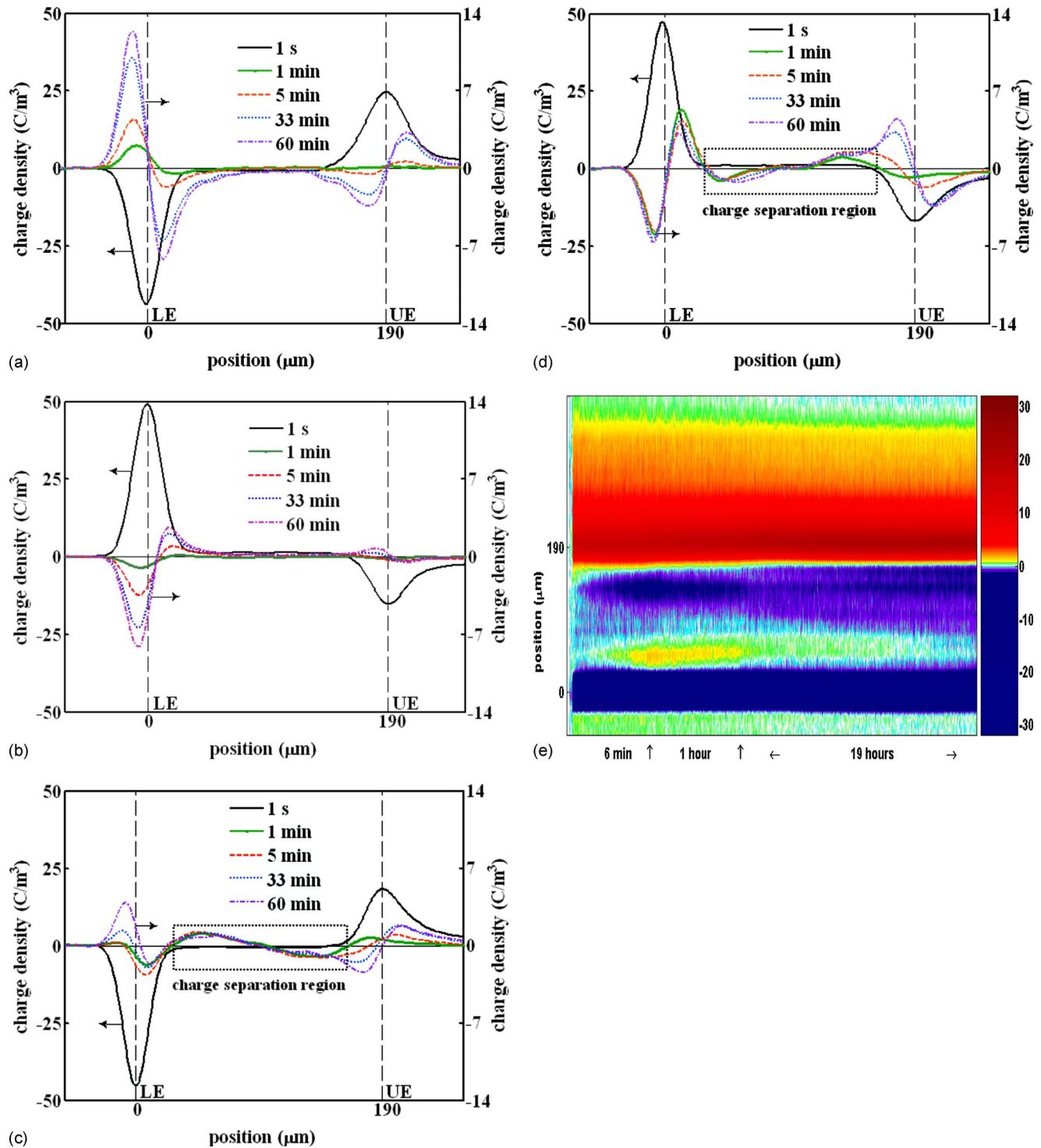


FIG. 2. (Color online) Space charge evolution in PPE under 50 kV_{dc}/mm at 25 °C for (a) and (b), at 50 °C for (c)–(e). For the sake of clarity, the profiles for times 1 min and beyond have been plotted as the difference between the profiles obtained at the respective times and that at 1 s. [(a), (c), and (e)] Positive voltage on UE. [(b) and (d)] Negative voltage on UE. (e) 3D representation of the profiles for 20 h stressing time. The time axis is linear in the three time intervals indicated. Scales are given to the right in C/m³ units.

polarity charges increasing synchronously in the following 16 min. Figure 2(e) shows a three-dimensional (3D)-plot of the time evolution of the profiles that enables one to appreciate the charge build up. A similar process (i.e., build up of a positive space charge region close to the LE and symmetric distribution of charge accumulating in the bulk at first and then slow build up of a positive space charge region at the

UE) occurs when a negative voltage is applied to the UE. Figure 2(d) shows the early features of space charge behavior for the long stressing time.

During the tests at 50 °C under different fields, we observed that the time for the appearance of the symmetric distribution of charge is dependent on the applied field and is independent on the voltage polarity. The field dependence of

TABLE I. Time for the onset of the appearance of symmetric distributing charge in PPE film under different field at 50 °C.

Applied field (kV/mm)	15	30	50	100
Time (s)	1800	340	30	0.3

the time for the appearance of the distribution is given in Table I. These times have been determined as the instant when the process can be distinguished adjacent to the LE. As the PEA technique has a limited spatial resolution and because of the overlapping of the image charge, it can be considered that the position at which the onset of symmetric distributing charge is best monitored is about 40 μm from the LE. Considering Fig. 2(e) as an example, negative charges, presumably injected from the LE, are dominant before the appearance of symmetric distributing charge.

Two possibilities can be envisaged for the origin of the symmetric distribution of charges.

- (i) Charges are injected from both electrodes and then transit across the bulk without being recombined. The accumulation at the opposite electrode would be due to some barrier-to-extraction effect. If the symmetric distributing charge formation were due to such process, this would imply that the charge injection from the two electrodes is symmetrical and that the transit time is the same for positive and negative carriers; otherwise one could not explain the symmetric build up of the distribution shown in Fig. 2(e). However, as mentioned above, the charge injection rate is asymmetrical from the two electrodes used in the study, and the drift of injected negative carriers appears faster than that of positive ones at 25 °C. This difference in mobility would be retained at 50 °C according to the trap-modulated mobility model.²²
- (ii) Under the applied field, pre-existing dissociated ion pairs are driven toward the electrodes. The symmetric distribution of charges might be generated by field-dependent charge separation and drift, which would be favored at the higher temperature (50 °C) and would explain why the behavior at 25 and 50 °C is that different. In this scenario, either one or the two charged centers could be mobile. The drift of these charged centers would lead to a depleted region at one of the electrodes and to an accumulation at the other electrode. Since mass transport is involved, the extraction of the species would not be possible and therefore the charge quantities detected would be symmetrical provided the material is initially macroscopically electrically neutral.

Very clearly, charge separation alone cannot explain all the features found in the profiles. Copious negative charge injection occurs at short time at the LE in Fig. 2(e) and carriers generated by this process become dominant at long stressing time. The time at which the ionic carriers accumulation is best resolved is from 10 to 60 min in Fig. 2(e). The accumulation process of ionic carriers superimposes on that of the electronic carriers injection and drift. The two pro-

cesses can be viewed as noninteracting as long as the distortion in the electric field profile remains negligible.

B. Space charge formation under ac stress

In order to investigate in more detail the charge separation behavior under dc stress at 50 °C, a low frequency square voltage stress (rms field of 50 $\text{kV}_{\text{ac}}/\text{mm}$) was applied to the PPE film. Considering the time for the appearance of charge separation under continuous dc stress, we selected frequencies of 5 and 100 mHz: the half periods are therefore, respectively, longer and shorter than the time for charge separation detection at that field and temperature.

Figure 3 shows the space charge profile evolution as a function of time for the two frequencies. The stressing time was 1 h in both cases. For the frequency of 5 mHz [Fig. 3(a)], the results have been obtained with continuous measurement, each profile being obtained from the averaging of acoustic signals over 60 pulses of the 1 kHz generator. In the 1 h time, this provides 18 periods of the HV stress. For the 100 mHz measurements [Fig. 3(b)], 200 profiles per period of the HV stress were produced, each of them being an average of the response to 60 pulses. The measurement takes about 20 s and represents actually more than one period of the HV stress. The procedure described in Ref. 5 was used for the purpose of data sorting and averaging. The representation is a concatenation of the results for each step. The results for the first and second periods are represented in an enlarged form for clarity.

As expected, the space charge behavior during the first half cycle (UE at positive voltage) for the first period at 5 mHz is similar to that under dc stress. After several seconds of the voltage application, the symmetric distribution of charges can be observed [Fig. 3(a)]. During the second half cycle (negative polarity on UE), the charge previously accumulated dissipates and reappears later on, forming again a symmetric distribution. As shown in Fig. 3(a), it is clear that the positive charge in the vicinity of LE shifts up to UE and the negative charge is seen in the vicinity of LE after 1/3 of this negative voltage half cycle. In these results, the maximum positive or negative charge density is of the order of 1 C/m^3 ; the probability for charges of opposite polarity to encounter and recombine during the migration through the insulation is small.

Along with the migration of the separated ionic species, there is probably also transport of injected electronic carriers, but the two processes cannot be distinguished easily. Considering the evolution of the pattern as a function of the stressing time, it can be observed that the bulk of the material tends to be charged more and more negatively. For example, the extent of the negatively charged region in front of the UE for the positive voltage half cycles grows in time. The extent of the positively charged region in front of the LE for the same half cycle correspondingly decreases in time. The imbalance between positive and negative injected carriers density in the course of time could explain the evolution toward negative charging in the bulk.

At long time, charge separation features tend to be less apparent. After 20 h stressing (results not shown), only nega-

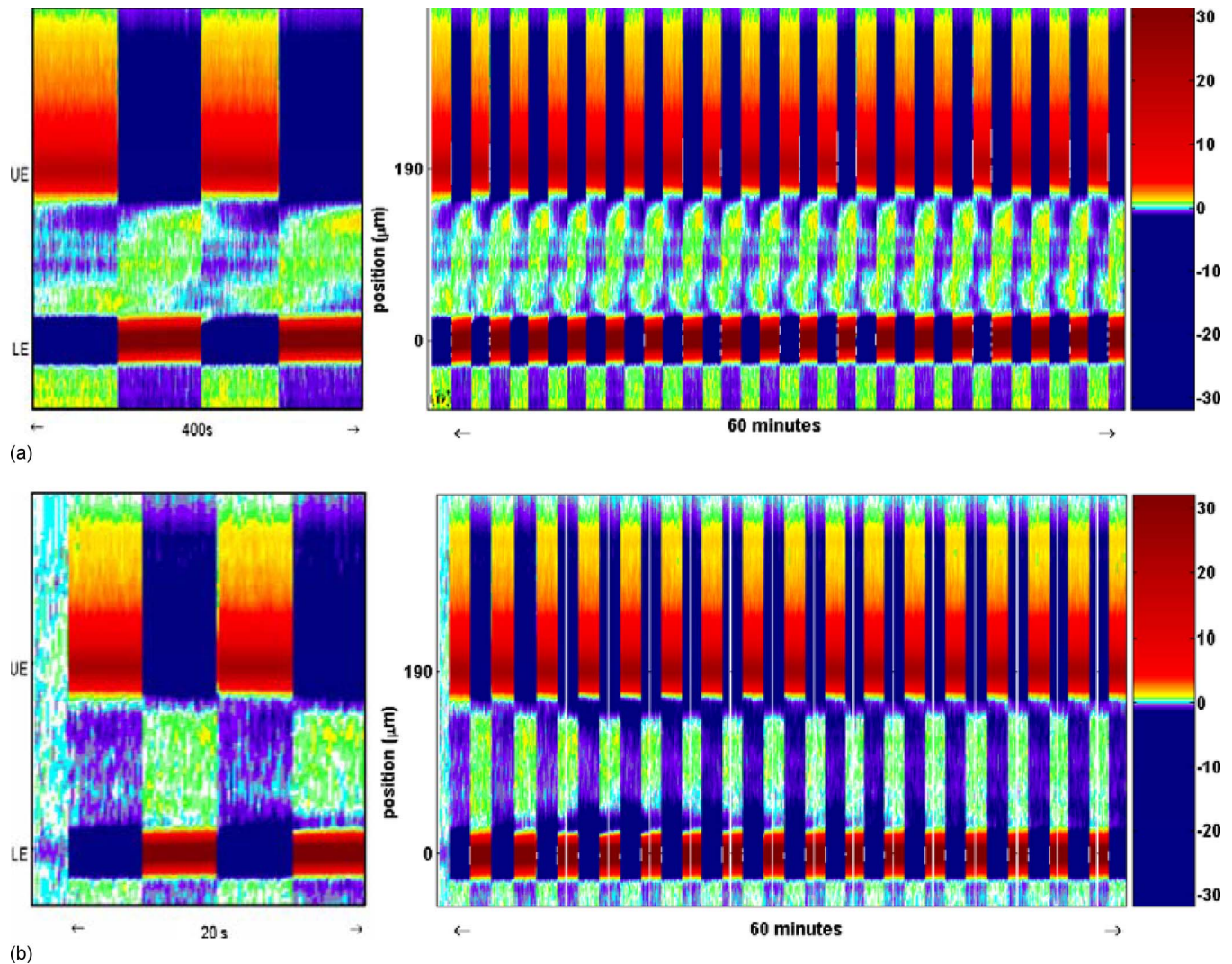


FIG. 3. (Color online) The space charge formation under ac square stress at 50 °C within 60 min (a) 5 mHz and (b) 0.1 Hz. To the left, the patterns for the first and second periods have been magnified.

tive (respectively, positive) charge can be detected in the volume for positive (respectively, negative) voltage half cycles. Possibly most of the dissociated charges have been neutralized, and the residual charge would be electronic carriers having a relatively high mobility. These charges would be generated preferentially at the LE whatever the polarity of the voltage. It can be also envisaged that the injection rate at the electrode increases with time.

When the frequency of the applied stress is 0.1 Hz, the existing dissociated ion pairs do not get enough time in a half cycle to be separated from each other before the next half cycle with opposite polarity voltage arrives. Therefore, the charge injected from the electrodes is dominant in the bulk for the first periods of the applied stress. Negative charge is dominant for positive half cycles and positive is dominant for negative half cycles, which is in agreement with the results presented in Figs. 2(a) and 2(b). The negative charge density close to the LE is at maximum in the fourth pattern (meaning time is about 15 min). It produces a negative heterocharge when the polarity is negative. In the same period of time, it can be observed also that the negative charge density tends to be higher at the UE. One could imagine that,

in a situation where positive and negative ionic species of different mobility coexist and are submitted to an alternating field, a segregation of the carriers occurs. In the present situation negative ions would be dominant close to the electrodes. However, if that were the case, one would expect also regions (bulk) with enhanced positive charge density and this is not apparently the case here. The growing up of this negative charge density adjacent to the electrodes can be understood on the basis of periodical injection of negative carriers. However, we have no explanation for the density decrease in time.

C. Charge migration

In order to further confirm the origin of the symmetric distribution of charges observed at 50 °C under dc stress, a two-layer sample, composed of two pieces of superposed PPE film, has been tested under 50 kV_{dc}/mm stress. The results show that the symmetric distribution can also be observed on both layers (Fig. 4). If the symmetric distributing charge was due to injected charge transiting across the insulation, positive and only positive charge should be located in

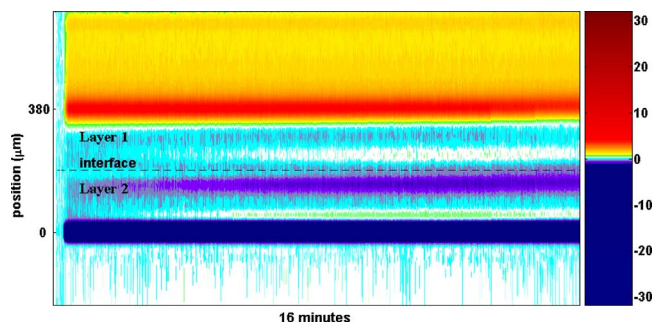


FIG. 4. (Color online) Space charge development within a two-layer PPE film during 16 min under 50 kV_{dc}/mm stress at 50 °C.

layer 2 and negative charge in layer 1 when the electronic carriers can easily go through the interface. However, this is inconsistent with the experimental results. This provides further evidence that the symmetric distribution of charges does not result from the transiting injected charge but rather from charge separation. The smaller charge density in layer 1 compared to layer 2 is due to the relative low amplitude resolution of the PEA method without amplitude correction. Bearing this in mind, this is useful information for the comparison of charge density in the vicinity of LE and UE.

Based on the above results, it can be concluded that the symmetric distribution of charges originates from the sample bulk. It can be due to charge separation, and the charge transport should be accordant with an ion transport mechanism. When HV is applied to the film, the separated charges move in an opposite way for positive and negative species. In order to detect this process with a space charge measurement technique, whatever it be, two conditions must be fulfilled: (i) the net charge must be non-null over a distance exceeding the spatial resolution of the method, and hence at least one of the charged species must have moved over that distance; (ii) the density of accumulated charge should be higher than the sensitivity limit of the method.

In addition, in the present situation the charged region forms starting from the electrodes and the image charge, along with injected carriers, overlap the separated charge build up, which therefore renders the detection more difficult. Supposing that the pairs of separated charges are initially uniformly distributed within the whole sample, we have considered that the separated charges can be detected in the profiles for an ion shifting distance of 40 μm on average. The ionic conductivity model²³ has been used to fit the data of the time of detection of the charges, i.e.,

$$v_d = a\nu_0 \exp\left[-\frac{\Delta}{k_B T}\right] 2 \sinh\left[\frac{Eae}{2k_B T}\right], \quad (1)$$

where v_d is the ion drift velocity, a is the hopping distance, ν_0 is the attempt to escape frequency (here $\nu_0(T=50\text{ °C}) \approx 2.7 \times 10^{13}$ Hz), k_B is the Boltzmann's constant, Δ is the activation energy, and E is the applied field. In this study, the ion drift velocity has been estimated as the shifting distance of 40 μm divided by the onset time for the symmetric charge separation detection given in Table I. With the appropriate parameters, the experimental data can be well fitted with the model as shown in Fig. 5. A hopping distance of 5.5

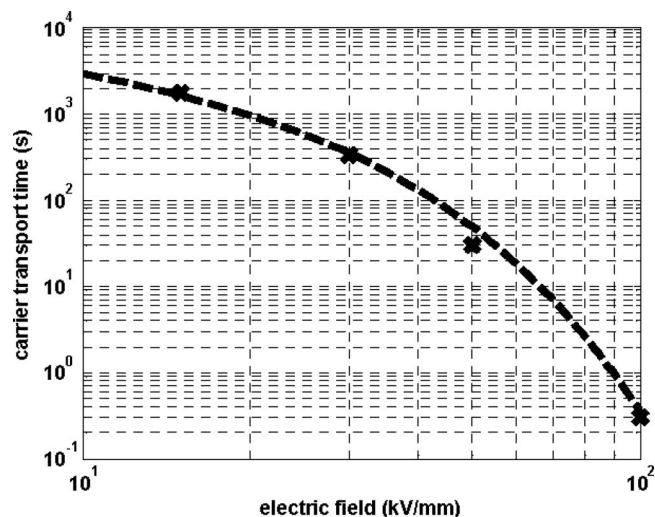


FIG. 5. Time for charge separation detection at 50 °C as a function of applied field (crosses) and fit to an ion transport model (dashed line).

nm and an activation energy of 0.86 eV have been deduced: these values appear as physically consistent for the model considered. This provides a further support that the formation of the symmetric distribution of charges results from the migration of ionic species.

Now that the bulk charge distribution phenomenon is considered as due to the migration of dissociated ions, one may wonder about the nature of the ions and what makes the significantly different charging behavior at 25 and 50 °C under 50 kV_{dc}/mm stress. Supposing that there are pre-existing ion pairs in the sample before HV application, according to the ionic conductivity mechanism and with the parameters obtained above, the symmetric distributing charge should be expected after 9 min of HV application at 25 °C. However, regardless of the polarity of applied voltage, only the charge injected from LE accumulates in the bulk during 10 h stressing at 25 °C. The recombination between the shifting dissociated ions and injected charge cannot be the cause. Indeed, otherwise the symmetrical charge separation could not occur at 50 °C either, because when the temperature increases, the injection is enhanced and so would be the probability of recombination events. Consequently, the symmetric charge distribution has to be attributed to charge separation. The required energy to separate charges has two sources: electrostatic field due to applied field and heat. For the same applied field, the dissociated ions may not get enough heat energy to be separated. This may be the reason of different charge behavior at 25 and 50 °C in this study. On the other hand, the charge separation shows a strong field dependency. At 25 °C, when the applied field increases up to 100 kV/mm, the symmetric distribution phenomenon can be detected again (Fig. 6). This suggests the occurrence of charge separation. The strong charge injection from LE under high applied field makes the density of charges in the vicinity of LE far lower than that near UE. This can also be observed under various applied fields at 50 °C (Fig. 7). It is clear that the charge density near both electrodes increases with applied field. The relative lower charge density in the vicinity of LE is due to the stronger injection from LE as mentioned earlier.

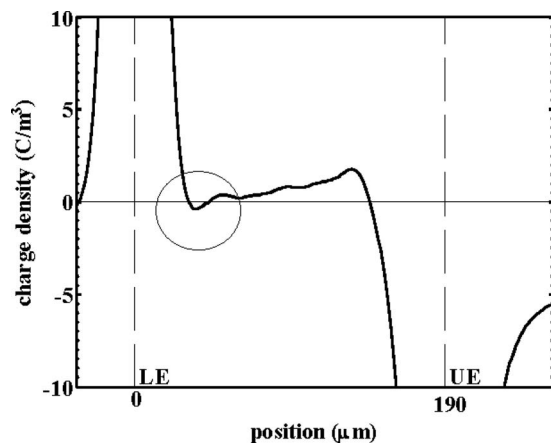


FIG. 6. Charge separation at 25 °C under 100 kV/mm. Stressing time: 17 min.

IV. CONCLUSIONS

The space charge behavior in a PPE copolymer at 25 and 50 °C under dc and ac stress has been investigated in this study. By combining measurements under dc and ac stress with different frequencies and applied field, and considering the time dependency of the profiles, it is shown that two independent processes are at play: homocharge formation at the LE due to electronic carriers injection and bulk generation of carriers providing a region with asymmetric distribution of charges in the bulk of the sample. The drift of injected carriers and their drift appear responsible for the slow build up of a heterocharge close to the UE.

The main conclusions are as follows.

- (1) The charge injection rate is determined by the electrode interfaces for a given temperature and applied field. The injection rate at the PPE/Al interface is higher than that at the PPE/semiconductor interface. This is independent of the polarity of the voltage applied to the electrodes in the PEA cell, i.e., this is true for both electrons and holes.
- (2) The charge separation is due to the presence of dissociated ions in the films. The charge separation rate de-

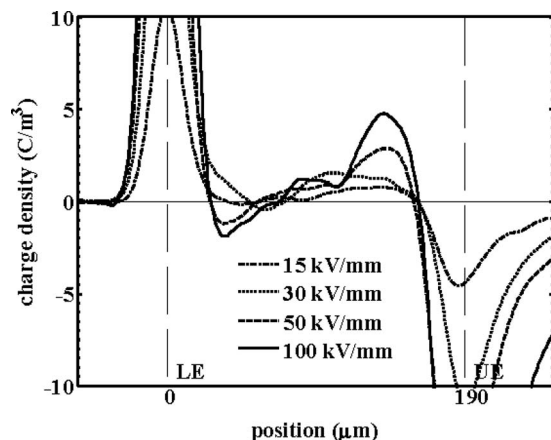


FIG. 7. Space charge profiles obtained at 50 °C under various fields. Profiles were taken at the time when the symmetric distribution of charges is the most prominent.

pends on the temperature and the applied field. It is significantly enhanced when increasing the temperature. The field dependence of the charge build up in the bulk appears consistent with an ion transport mechanism.

- (3) The frequency of the applied ac field has a great impact on the space charge behavior. In this study, only injected charges can be observed at 0.1 Hz and both injected charges and dissociated charges at 5 mHz. Therefore, measurements under ac stress can constitute a way of discriminating bulk charge generation and carriers provided by injection.

ACKNOWLEDGMENTS

We acknowledge the French Ministry of education for providing a Post-Doc grant. Two of the authors (F.Zheng and Y.Zhang) are grateful to the National Natural Science Foundation of China (Contract No. 50537040), Shanghai Committee of Science and Technology (Contract No. 07DZ22302) and Program for Young Excellent Talents in Tongji University for financial support.

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